

Code 2 (Separate from code 1)

Suchanun Oupkaew

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```
library(dplyr)
```

```
##  
## Attaching package: 'dplyr'  
  
## The following objects are masked from 'package:stats':  
##  
## filter, lag  
  
## The following objects are masked from 'package:base':  
##  
## intersect, setdiff, setequal, union
```

```
library(ggplot2)
```

```
credit.approval = read.csv("/Users/cream/Downloads/archive/clean_dataset.csv")  
credit.approval <- subset(credit.approval, select = c('Gender', 'Age', 'Married', 'Ethnicity', 'YearsEmployed'))  
  
#making a function that creates bar with count of approved for each bar  
plot_bar = function(df, x, y, response, x_title, y_title, legend_title, title) {  
  
  ggplot(df, aes(x = x, y = y, fill = response)) +  
    geom_bar(stat = 'identity', position = 'dodge') +  
    geom_text(aes(label = y), position = position_dodge(width = 0.9), vjust = -0.5) +  
    scale_fill_discrete(name = as.character(legend_title)) +  
    labs(  
      title = as.character(title),  
      x = as.character(x_title),  
      y = as.character(y_title)  
    )  
}
```

```
#fit the logistic regression
```

```
logistic = glm(Approved ~ Gender + Age + Married + Ethnicity + YearsEmployed + Employed + CreditScore, data = credit.approval,  
summary(logistic)
```

```
##  
## Call:
```

```
## glm(formula = Approved ~ Gender + Age + Married + Ethnicity +
##     YearsEmployed + Employed + CreditScore, family = binomial,
##     data = credit.approval)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -2.548885   0.527385  -4.833 1.34e-06 ***
## Gender        -0.115073   0.212482  -0.542 0.588117
## Age           0.016520   0.009549   1.730 0.083626 .
## Married       0.505024   0.232849   2.169 0.030091 *
## EthnicityBlack 0.899725   0.400673   2.246 0.024734 *
## EthnicityLatino -1.899073  0.619446  -3.066 0.002171 **
## EthnicityOther 0.569786   0.611532   0.932 0.351473
## EthnicityWhite 0.337960   0.362163   0.933 0.350732
## YearsEmployed 0.171236   0.041663   4.110 3.96e-05 ***
## Employed      0.981213   0.264187   3.714 0.000204 ***
## CreditScore   0.230634   0.051286   4.497 6.89e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 948.16  on 689  degrees of freedom
## Residual deviance: 669.51  on 679  degrees of freedom
## AIC: 691.51
##
## Number of Fisher Scoring iterations: 6
```

Since logistic shows that only certain predictors are significant, so I need to make a new data set where I remove all insignificant predictors

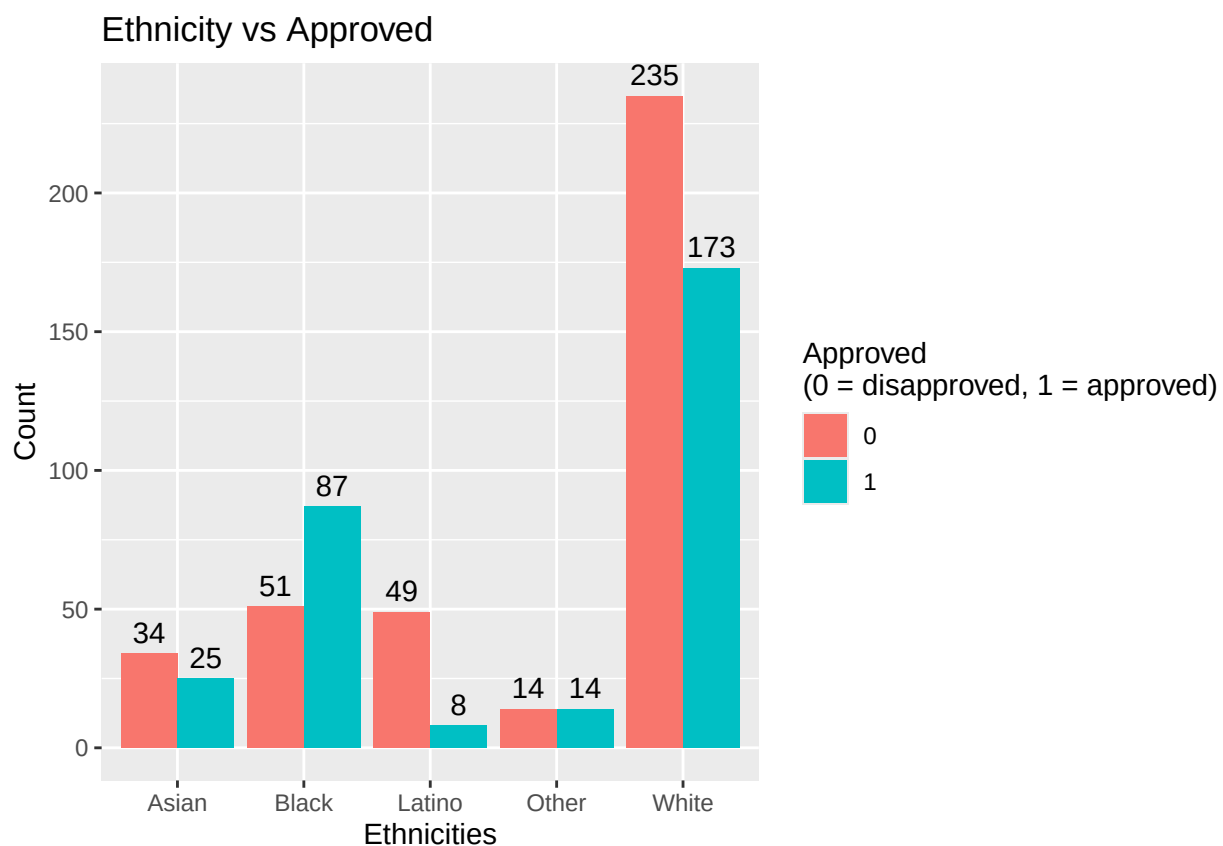
```
new_data = subset(credit.approval, select = c('Married', 'Ethnicity', 'YearsEmployed', 'Employed', 'CreditScore'))
new_data = new_data %>%
  filter((Ethnicity != "Other") & (Ethnicity != "White"))
new.fit = glm(Approved ~ Married + Ethnicity + YearsEmployed + Employed + CreditScore, data = new_data,
summary(new.fit))
```

```
##
## Call:
## glm(formula = Approved ~ Married + Ethnicity + YearsEmployed +
##     Employed + CreditScore, family = "binomial", data = new_data)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -2.44295   0.51414  -4.752 2.02e-06 ***
## Married       0.96866   0.40171   2.411 0.01590 *
## EthnicityBlack 0.89173   0.39891   2.235 0.02539 *
## EthnicityLatino -1.69103  0.61077  -2.769 0.00563 **
## YearsEmployed 0.17798   0.05765   3.087 0.00202 **
## Employed      0.92949   0.46152   2.014 0.04401 *
## CreditScore   0.22580   0.07716   2.926 0.00343 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## (Dispersion parameter for binomial family taken to be 1)
##
## Null deviance: 351.35 on 253 degrees of freedom
## Residual deviance: 220.09 on 247 degrees of freedom
## AIC: 234.09
##
## Number of Fisher Scoring iterations: 5
```

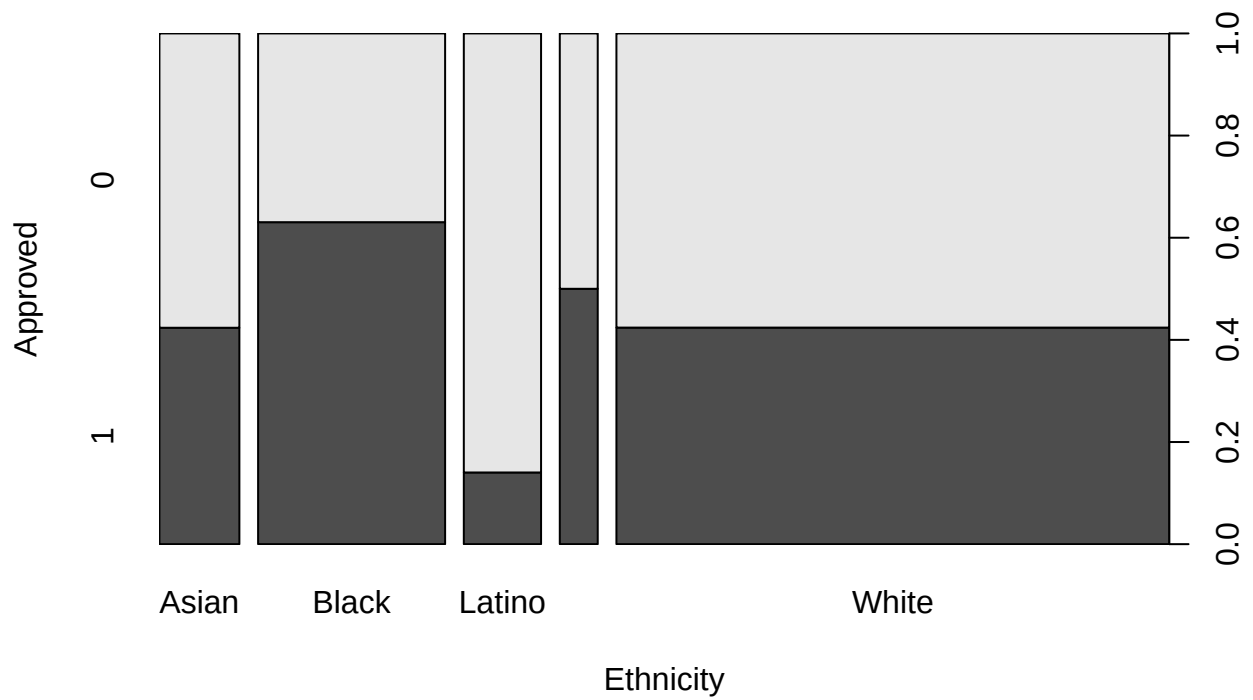
```
#For Ethnicity
```

```
credit.approval$Approved <- factor(credit.approval$Approved)
ethnic = credit.approval %>%
  group_by(Ethnicity, Approved) %>%
  summarise(amount = n(), .groups = 'drop')
plot_bar(ethnic, ethnic$Ethnicity, ethnic$amount, ethnic$Approved, 'Ethnicities', 'Count', 'Approved \n
```



```
credit.approval$Approved <- as.factor(credit.approval$Approved)
credit.approval$Married <- as.numeric(credit.approval$Married)
credit.approval$Ethnicity <- factor(credit.approval$Ethnicity)
ethnis.fit = glm(Approved ~ Ethnicity, data = credit.approval, family = "binomial")
eth = data.frame(Ethnicity = rep(c("Black", "Latino"), 100))
eth$Approved = predict(ethnis.fit, eth, type = 'response')

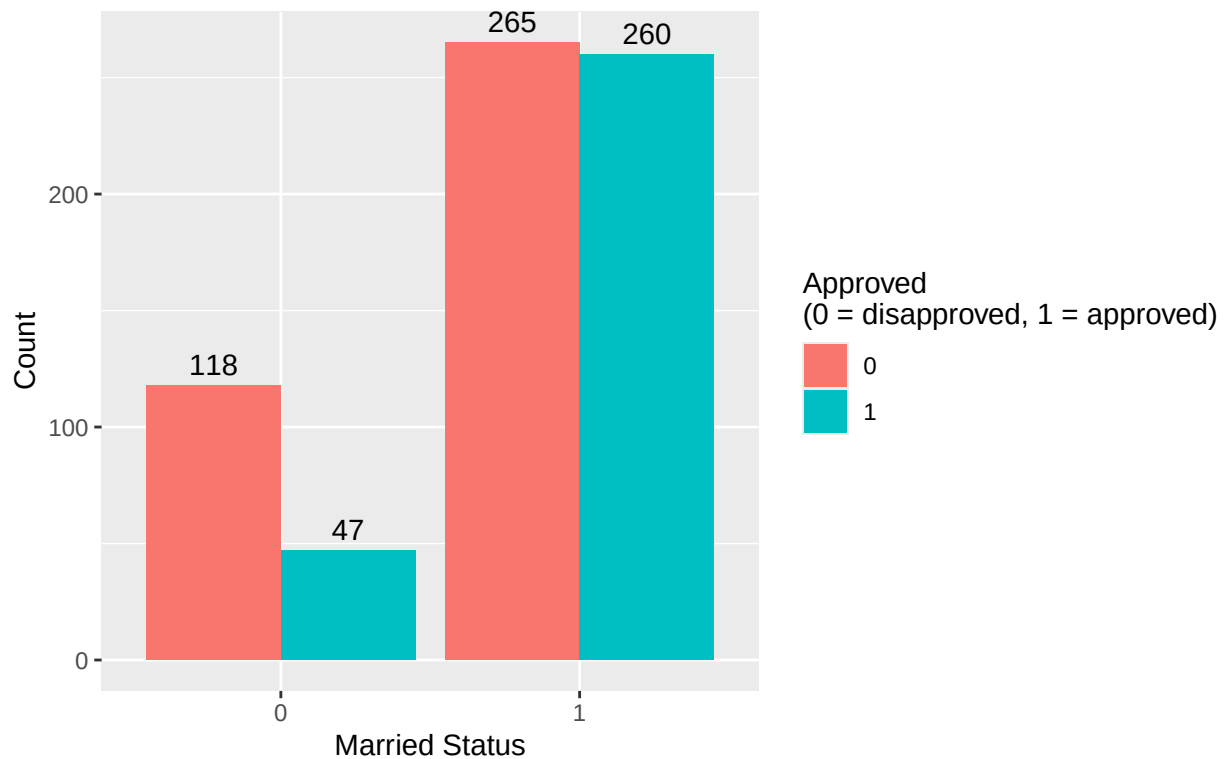
plot(Approved ~ Ethnicity, data = credit.approval)
```



#for Married

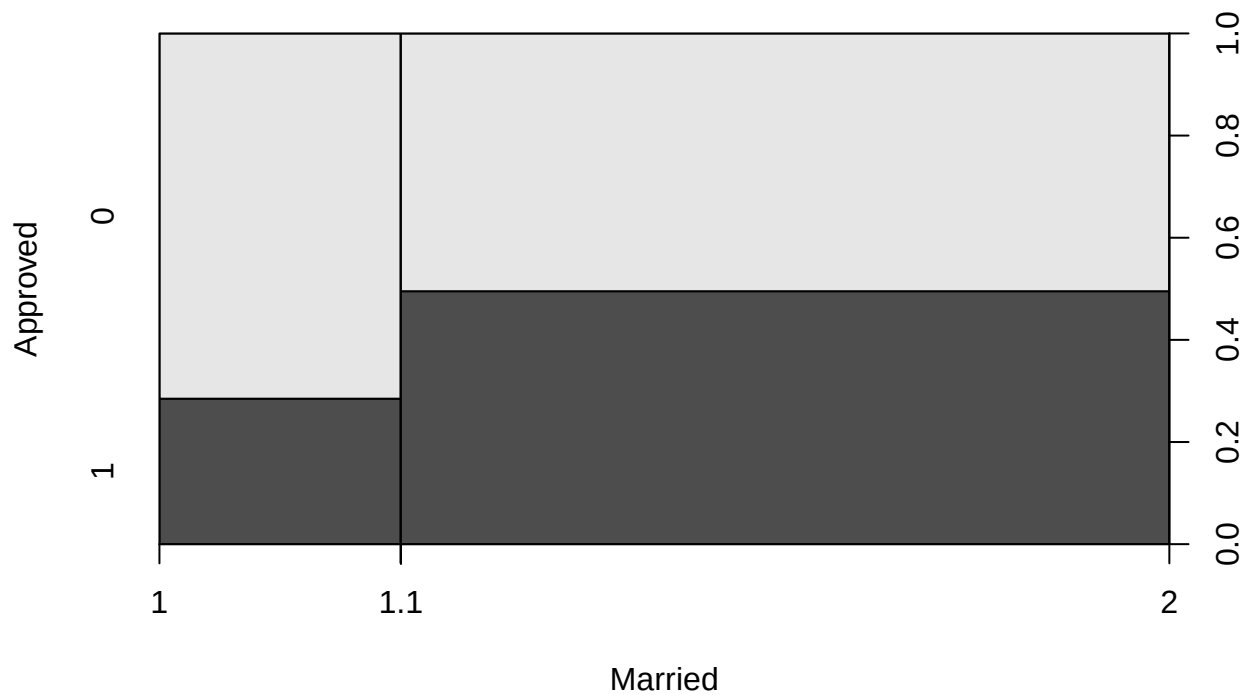
```
credit.approval$Married <- factor(credit.approval$Married)
countapproved = credit.approval |>
  group_by(Married, Approved) |>
  summarise(Count = n(), .groups = 'drop')
plot_bar(countapproved, countapproved$Married, countapproved$Count, countapproved$Approved, "Married St.
```

Married vs Approved (1 = not married, 2 = married)



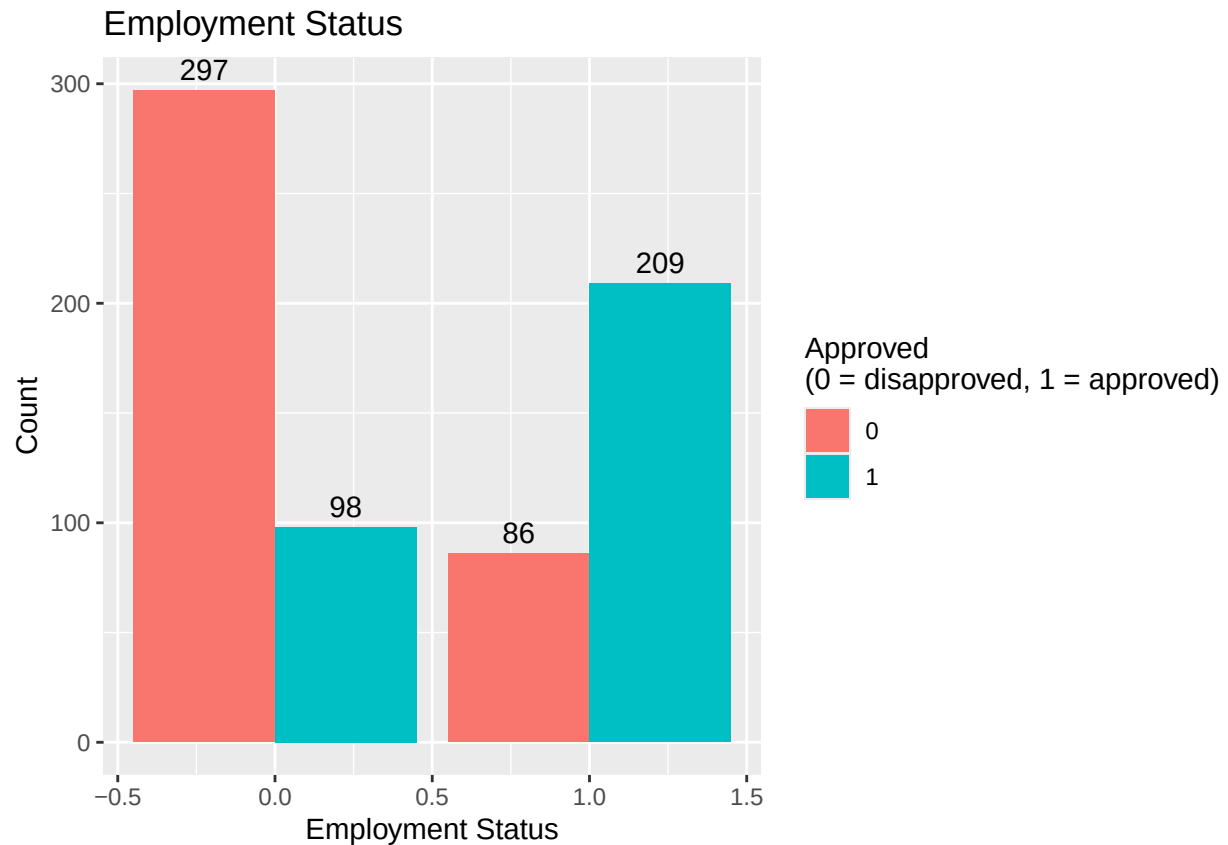
I try to plot logistic regression but both axes are categorical values

```
credit.approval$Married <- as.factor(credit.approval$Married)
credit.approval$Approved <- as.factor(credit.approval$Approved)
credit.approval$Married <- as.numeric(credit.approval$Married)
mar.fit = glm(Approved ~ Married, data = credit.approval, family = 'binomial')
pr = data.frame(Married = c(0, 1))
pr$Approved = predict(mar.fit, pr, type = 'response')
plot(Approved ~ Married, data = credit.approval)
```



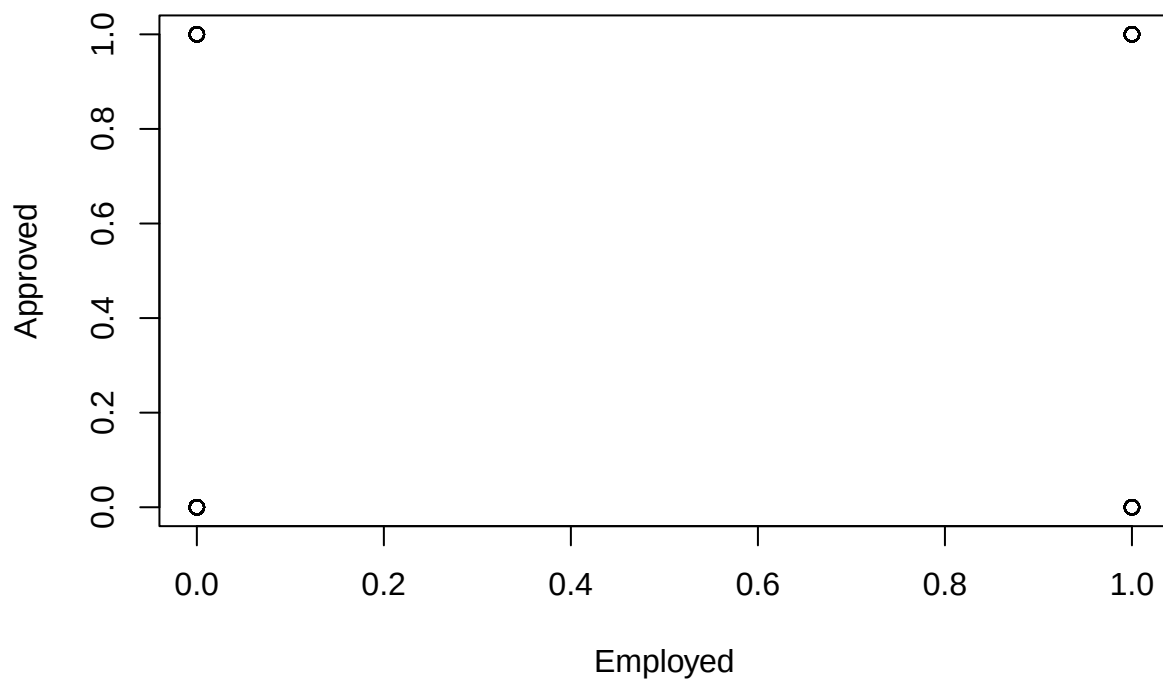
#For Employed

```
employed = credit.approval %>% #this data is for counting
  group_by(Employed, Approved) %>%
  summarise(Sum = n(), .groups = 'drop')
plot_bar(employed, employed$Employed, employed$Sum, employed$Approved, "Employment Status", "Count", 'A
```



```
#mar.fit = glm(Approved ~ Married, data = credit.approval, family = 'binomial')
#pr = data.frame(Married = c(0, 1))
#pr$Approved = predict(mar.fit, pr, type = 'response')
#plot(Approved ~ Married, data = credit.approval)

empl.fit = glm(Approved ~ Employed, data = credit.approval, family = "binomial")
empl = data.frame(Employed = c(0,1))
empl$Approved <- predict(empl.fit, empl, type = 'response')
plot(Approved ~ Employed, data = new_data)
```



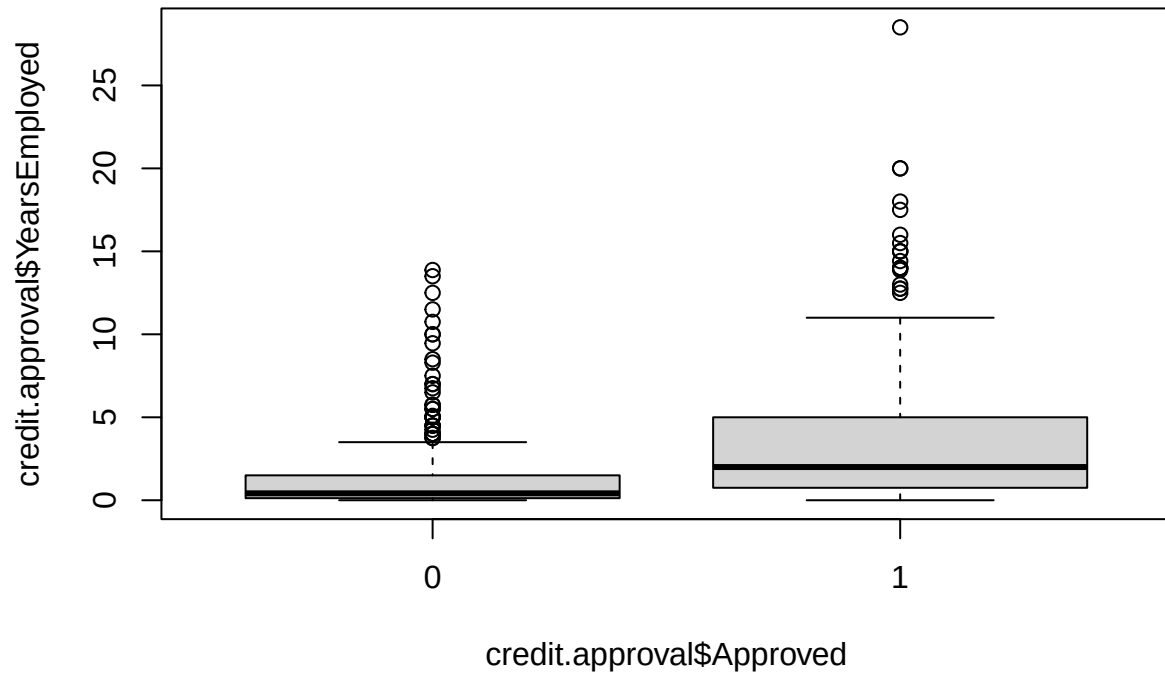
```
#predict using same model with new data
pred.data = tibble(
  Married = c(1, 1, 1,1), #if all applicant are all married
  Ethnicity = c('Black', 'Black', 'Black', 'Black'), #all black
  YearsEmployed = c(2,3,4,5),
  CreditScore = c(2,3,4,5),
  Employed = c(1,1,1,1), # all are employed
)
predict(new.fit, pred.data, type = 'response')
```

```
##           1           2           3           4
## 0.7603346 0.8261098 0.8767593 0.9141883
```

```
#For YearsEmployed
```

```
boxplot(credit.approval$YearsEmployed ~ credit.approval$Approved,
  main = "YearsEmployed ~ Approval Status")
```


YearsEmployed ~ Approval Status



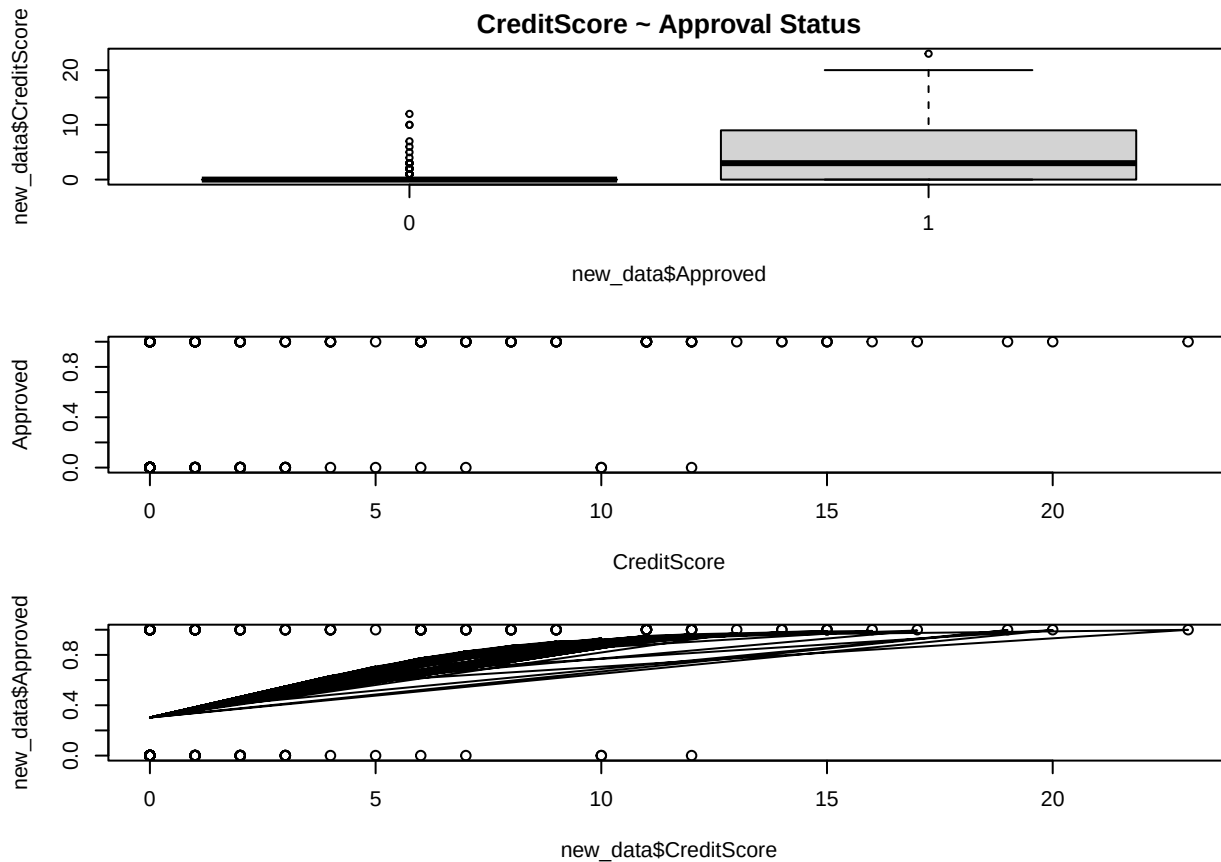
```
yrs.fit = glm(Approved ~ YearsEmployed, data = credit.approval, family = binomial(link = "logit"))  
plot(Approved ~ YearsEmployed, data = credit.approval)
```



#For CreditScore

```
par(mfrow = c(3, 1), mar = c(4, 4, 2, 1))
boxplot(new_data$CreditScore ~ new_data$Approved,
        main = "CreditScore ~ Approval Status")
credit.fit = glm(Approved ~ CreditScore, data = new_data, family = binomial(link = 'logit'))

plot(Approved ~ CreditScore, data = new_data)
plot(new_data$CreditScore, new_data$Approved)
lines(new_data$CreditScore, credit.fit$fitted.values) #unable to graph logistic reg. line
```



#For Job Industry Analysis: Is there a significant background factor (jobs) that affects the credit approval?

```
job = read.csv("/Users/cream/Downloads/archive/clean_dataset.csv")
fit.job = glm(Approved ~ Industry, data = job, family = 'binomial')
jobsum = summary(fit.job)
```

- It shows that Healthcare, IT, Material, Utilities are significant industries for Approved. To have a better estimation, we can remove the insignificant predictors.

```
job.new = job |>
  filter(Industry %in% c("Healthcare", "InformationTechnology", "Materials", "Utilities"))
job.new$Industry = as.factor(job.new$Industry)
jobnew2 = subset(job.new, select = c(Industry, Approved))
#fit logistic regression
fit.job2 = glm(Approved ~ Industry, data = jobnew2, family = 'binomial')
summary(fit.job2)
```

```
##
## Call:
## glm(formula = Approved ~ Industry, family = "binomial", data = jobnew2)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -1.8827     0.4057  -4.641 3.47e-06 ***
## IndustryInformationTechnology  2.7651     0.5314   5.203 1.96e-07 ***
```

```
## IndustryMaterials          2.5187      0.4704      5.355 8.56e-08 ***
## IndustryUtilities          3.5567      0.6021      5.907 3.48e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 287.38  on 209  degrees of freedom
## Residual deviance: 224.72  on 206  degrees of freedom
## AIC: 232.72
##
## Number of Fisher Scoring iterations: 4
```

```
df.indd = sample(c("InformationTechnology", "Materials", "Utilities", "Healthcare"), 10, replace = T)
df.indd
```

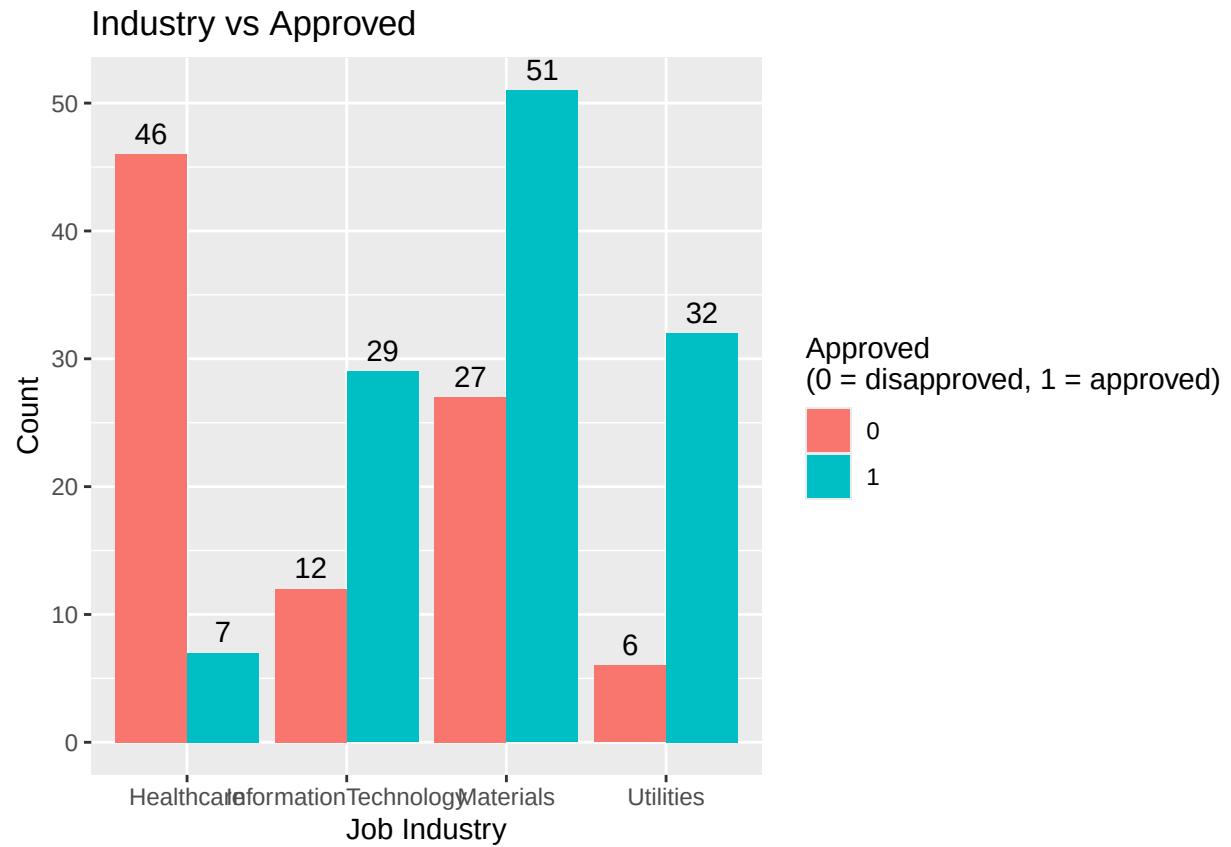
```
## [1] "Healthcare"      "InformationTechnology" "Healthcare"
## [4] "Utilities"        "InformationTechnology" "InformationTechnology"
## [7] "Healthcare"      "Utilities"           "Healthcare"
## [10] "Materials"
```

```
pred.job = data.frame(Industry = df.indd)
predict(fit.job2, pred.job, type = 'response') #probability of getting approved for each industry
```

```
##          1          2          3          4          5          6          7          8
## 0.1320755 0.7073171 0.1320755 0.8421053 0.7073171 0.7073171 0.1320755 0.8421053
##          9          10
## 0.1320755 0.6538462
```

```
#count the approved for each industry
count.job = job.new |>
  group_by(Industry, Approved) |>
  summarise(num = n(), .groups = 'drop')
```

```
count.job$Approved <- as.factor(count.job$Approved)
plot_bar(count.job, count.job$Industry, count.job$num, count.job$Approved, "Job Industry", "Count", "Approved")
```



- Healthcare is being treated as a reference category